The Institution's Weight: A Starter for the Extension

Companion note to the manuscript

15 October 2026

1 The question

In the manuscript the rule's weight on the base component, μ, is given. Its motivation is the author's account of information asymmetry and institutional development (14 October 2026): a credential buys a cleaner status than the disorganised status that comes from unverified providers, and that is why status becomes important. An institution allocating positions cannot see the quality it cares about. What it can read without verifying it is disorganised: claims, reputations and signals from providers it does not check. A credential is organised status: standardised, issued under rules, and cheap to check. The weight $1-\mu$ is how far the rule leans on that organised status, and μ how far it still reads the rest. As credentialing becomes organised, an institution has reason to lean on it more, which is the shift from disorganised to organised status in the story of institutional development. Once the credential can be bought, its holders are those who can afford it, which is what the manuscript studies. The extension asks why an institution sets μ where it does. Its answer has two sides. The cost side, what a higher μ does to who pays and what they take, is in the manuscript. The benefit side, what a higher μ does to the quality of the winner, is not.

2 What the manuscript already gives

Statement	Content	Rests on
The gain falls with μ	With the incumbent scored as an entrant, $\Delta(m;\mu)$ does not rise in μ at any m.	M28, K8
The count falls with μ	With costs fixed, k^* does not rise in μ.	M28, K8
The entrants' share falls with μ	The probability that the prize goes to an entrant does not rise in μ.	M31, K7
The benchmark at $\mu \to 0$	$\Delta(m) \to V/(m+2)$ for every Q.	M9
Entry buys nothing at $\mu = 1$	At $\mu = 1$ the score is r for every player, so $F = G$, $\varphi = 0$ and $\Delta(m) = 0$; with $c > 0$ and u strictly increasing, $\kappa > 0$ and no challenger enters.	One step from M1; `no_entry_at_one`

So, with the fee, the prize and the wealth profile fixed, a rule that weighs the base component more draws no more entrants and gives them no larger share of the prize, and at $\mu = 1$ it draws none.

3 What is missing: the benefit side

Let r_W be the base component of the winner. The institution's benefit from a higher μ is a better r_W. Two endpoints follow from the score in one step each, both proved in `lean/mathlib/WeightEndpoints.lean`; what lies between them is settled in `lean/mathlib/WeightSelection.lean` (M32).

- **W-1 (proved).** At $\mu = 1$ no challenger enters, every score is r, and the winner is the player with the highest base component, so r_W is the maximum of the $Q+1$ draws of r (`winner_base_eq_sup_at_one`).

- **W-2 (proved).** At $\mu = 0$ every score is $s_i e_i$, so the winner is picked by a rule on the bought components alone; with r independent of s and of a common law, r_W has the law of r (`score_at_zero`, `winner_base_law`). The incumbent, scored as an entrant, always holds the bought component, so the winner is defined.

Between the endpoints two forces act. A higher μ weighs r more in every score, which helps selection on r: with the entrants fixed, the winner's base component does not fall (M32). It also lowers entry (M28), which moves some challengers from the entrant law F to the outsider law G, and that can hurt: a common bonus washes out when every challenger holds it, and favours its holders once some stop buying.

- **W-3 (settled, M32).** $\mathbb{E}[r_W]$ does not fall in μ with the entrants fixed, pointwise and for any laws (`winner_base_mono`). With entry responding it can fall. With one challenger, r uniform, a bonus $s \equiv 1$ and $V = 1$, the gain is $\frac{1}{2}$ at $\mu = \frac{1}{2}$ and $\frac{3}{8}$ at $\mu = \frac{2}{3}$; a challenger with $\kappa = \frac{7}{16}$ enters at the first and not at the second, and $\mathbb{E}[r_W]$ is strictly lower at the second (`selection_falls_with_exit`). The conjecture as stated is false.

4 The institution's objective, as a construct

A candidate objective is

$$W(\mu) = \mathbb{E}[r_W](\mu) + \alpha\, c\, k^*(\mu) - \nu\, \gamma(\mu),$$

with $\alpha \geq 0$ the value to the institution of a fee collected, γ a cost of reading the unverified part of the score, which rises in μ, and $\nu \geq 0$ its price. In the terms of Section 1, ν is how disorganised the uncredentialed information is. Each term is a construct. The recorded datum is the rule an institution publishes: Chiba City's four points for paid care against the rest of its index, or the points an Italian contracting authority gives a certificate. The observer orders published rules by μ without measuring it (measurement map, the μ row), and W is admissible only as an ordering of μ that rationalises those choices.

The questions the extension would answer:

- **W-4.** When is the optimal μ interior? At $\mu = 1$ the fee term is zero (no entrant) and verification is dearest; near $\mu = 0$ the winner is no better than a draw of r (W-2).
- **W-5.** Does a change of the wealth profile move the optimal μ? The count responds to a single-crossing widening by the sign rule of K2 (M13), so the fee term moves with the widening, in a direction set by the cut-off wealth against the break-even wealth.
- **W-6.** Does better unverified information raise the optimal μ? A fall in ν lowers the cost of weighing the base component. With the entrants fixed the benefit side does not fall in μ (M32), but with entry responding it need not rise, so the answer turns on how entry moves.
- **W-7.** Institutional development. As credentialing becomes organised, so that the credential tracks what the institution values more closely, does the optimal μ fall? In the model the credential's content is the law of s; a credential that is cleaner in this sense would be one whose s is less noisy. That is not yet a primitive the model varies.
- **W-8.** When does the institution choose closure, $\mu = 0$? Only when the fee term and the saving on verification outweigh the loss in r_W from W-1 to W-2.

5 Literature to read first

The application search of 14 October 2026 found no paper in which a selector chooses the weight on a purchasable credential given a cost of verification; that is a statement about the abstracts read (`LITERATURE.tex`, the application search of 14 October 2026), not yet a novelty claim. The nearest models, by what they share with the extension:

- Frankel and Kartik (2022): a decision maker commits to under-using data that agents can manipulate. The weight is chosen; the cost of gaming is continuous and there is no fee. **[A]**.
- Hancart (2026): firms choose the accuracy and difficulty of a test in competition for candidates. The test is chosen; candidates differ only in productivity. Read in Sections 1 to 4. **[A]**.
- Liu et al. (2022), Appendix B.2: with a skill the school cannot measure, some randomization of the reward can be optimal for the school. Read in full (L32).
- Ben-Porath et al. (2014): a principal allocates one object and can verify a report at a cost. Verification is chosen per agent; there is no credential. **[A]**.
- Stahl and Strausz (2017) and Lizzeri (1999): certification as a market, chosen by sellers, buyers or an intermediary. **[A]**.

Each would need a full read, a record and Lean before the extension cites it for positioning.

6 What this gives the manuscript

One paragraph in Appendix D of the manuscript ("The institution's weight"), resting only on the table of Section 2: the model gives the cost side of the institution's choice of μ and not the benefit side, and the choice itself is left to the extension.

References

Elchanan Ben-Porath, Eddie Dekel, and Barton L. Lipman. Optimal allocation with costly verification. *American Economic Review*, 104(12):3779–3813, 2014. doi: 10.1257/aer.104.12.3779.

Alex Frankel and Navin Kartik. Improving information from manipulable data. *Journal of the European Economic Association*, 20(1):79–115, 2022. doi: 10.1093/jeea/jvab017.

Nathan Hancart. Selection procedures in competitive admission. Preprint 2510.12653, arXiv, 2026. Version 4, June 2026.

Lydia T. Liu, Nikhil Garg, and Christian Borgs. Strategic ranking. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 151 of *Proceedings of Machine Learning Research*, pages 2489–2518, 2022. arXiv:2109.08240.

Alessandro Lizzeri. Information revelation and certification intermediaries. *RAND Journal of Economics*, 30 (2):214–231, 1999. doi: 10.2307/2556078.

Konrad Stahl and Roland Strausz. Certification and market transparency. Discussion paper, open access lmu, Ludwig-Maximilians-Universität München, 2017.